1D CNN — Architecture Variant Comparison

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Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic
Total trials	360 (train pool 324, hold-out 36)
Classes	6: Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap
Sensor channels	20 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	per_trial_zscore
CV folds · shuffle	5 · True
Augmentation	on · copies=6 · time_shift, time_warp, time_mask, amplitude_scale, baseline_offset, channel_dropout
Epochs · batch size	60 · 16

Sensor grid (✓ enabled, ✗ disabled, — not present)

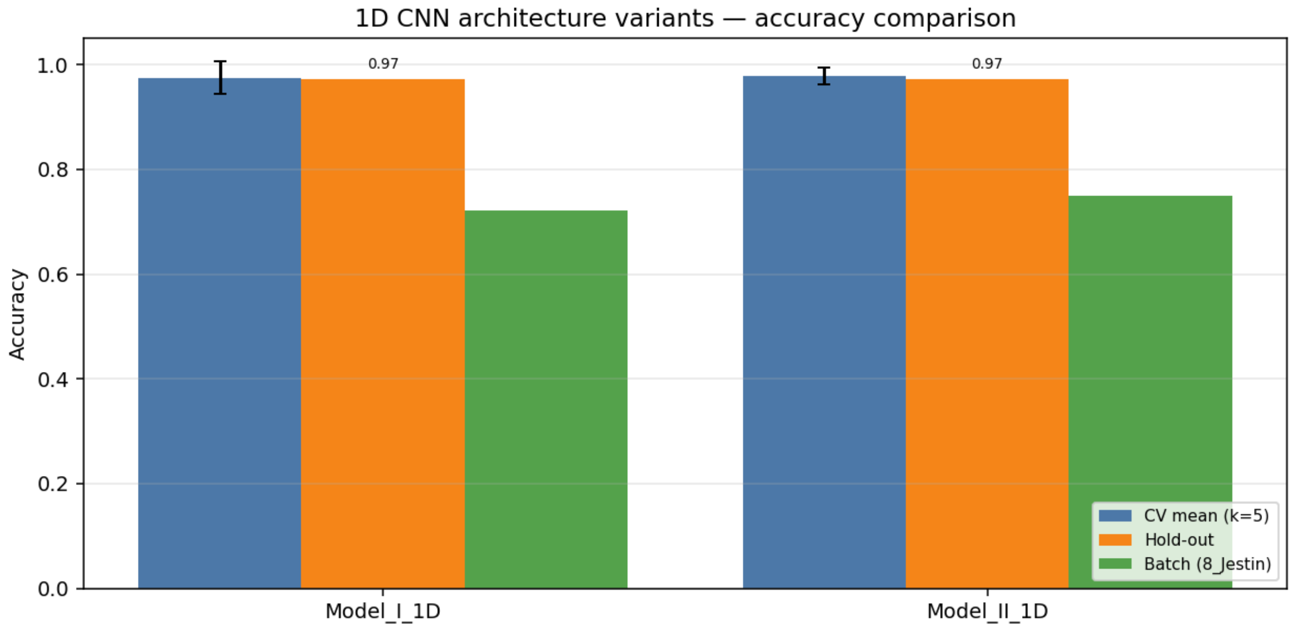
Hand / Segment	IMU_mid	IMU_prox	Flex_MCP	Flex_PIP
left palm	✓	—	—	—
left thumb	✓	✓	✓	✓
left index	✓	✓	✓	✓
left middle	✓	✓	✓	✓
left ring	✓	✓	✓	✓
left pinky	✓	✓	✓	✓
left wrist	✓	—	—	—
right palm	✓	—	—	—
right thumb	✓	✓	✓	✓
right index	✓	✓	✓	✓
right middle	✓	✓	✓	✓
right ring	✓	✓	✓	✓
right pinky	✓	✓	✓	✓
right wrist	✓	—	—	—

Enabled: 44 · Disabled: 0 · Modalities: Flex → 20 channels total.

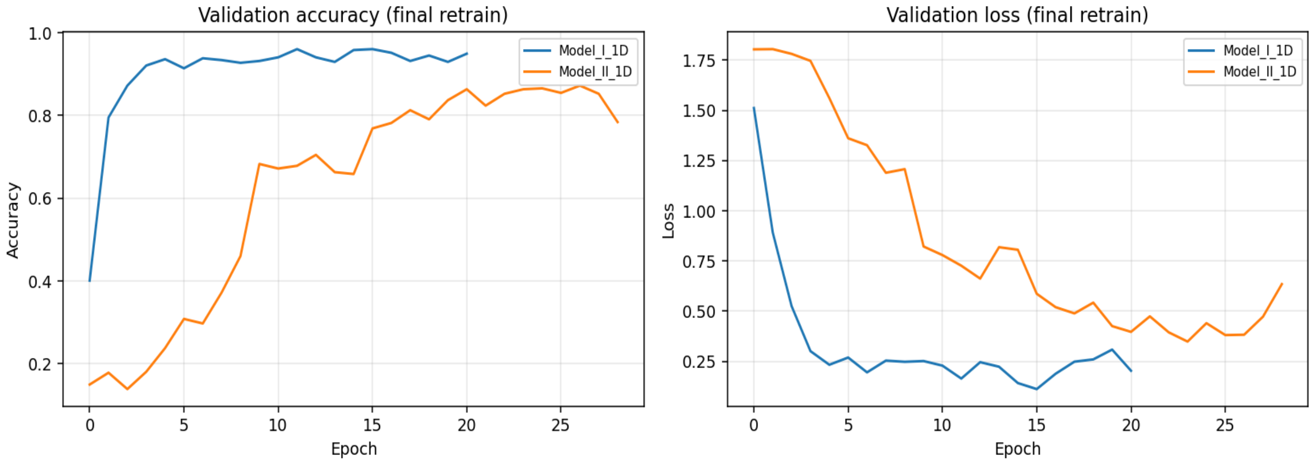
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Model_I_1D	118,630	0.9752	0.0304	0.9722	0.7461	0.7222	26/36
Model_II_1D	73,926	0.9784	0.0157	0.9722	0.1889	0.7500	27/36

Best on hold-out: **Model_I_1D** · Best on CV: **Model_II_1D**



Training curves (final retrain on full training pool)



Validation accuracy (final retrain)
Accuracy
Epoch
Model_I_1D
Model_II_1D
Validation loss (final retrain)
Loss
Epoch
Model_I_1D
Model_II_1D

Variant: Model_I_1D

1-D adaptation of published Model I:
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

Trainable params	118,630
CV mean ± std	0.9752 ± 0.0304
Hold-out test acc / loss	0.9722 · 0.7461
Batch-test acc	0.7222 (26/36)

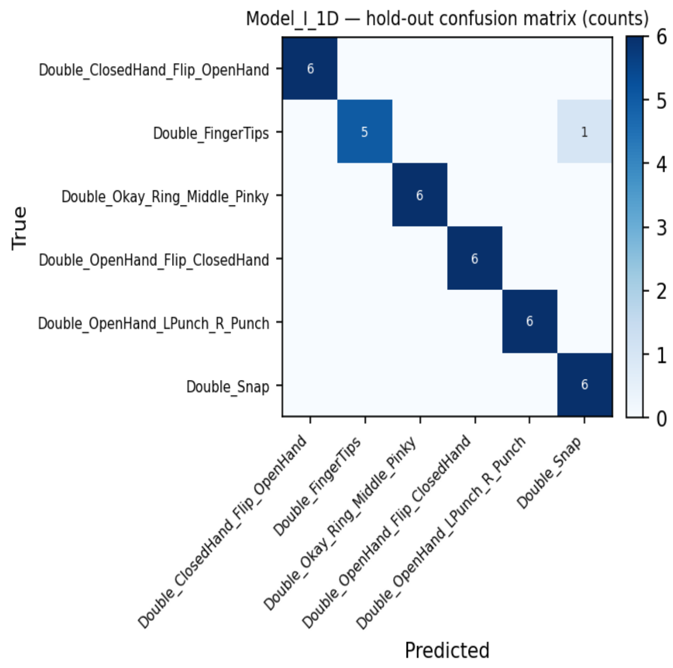
Cross-validation folds

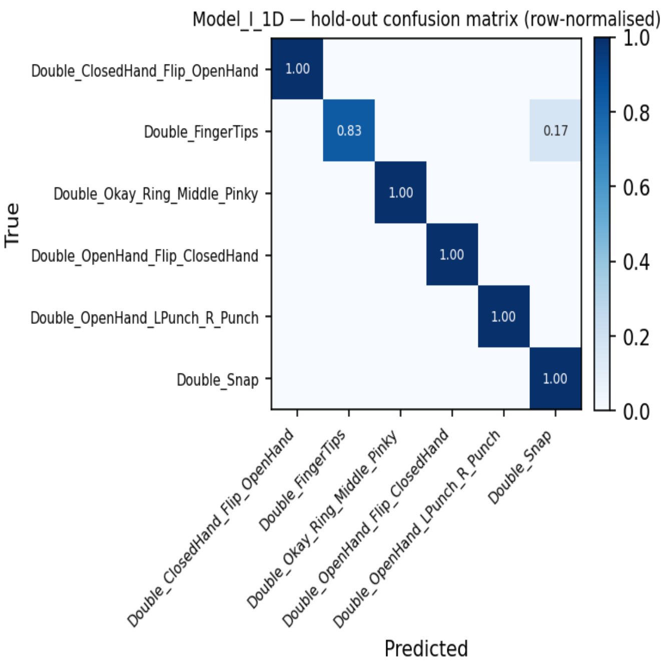
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1813	65	1.0000	1.0000	1.0000
2	1813	65	0.9385	0.9692	0.9692
3	1813	65	1.0000	1.0000	1.0000
4	1813	65	1.0000	1.0000	1.0000
5	1820	64	0.9375	0.9531	0.9375

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	0.833	0.909	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	1.000	1.000	1.000	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	0.857	1.000	0.923	6
macro avg	*0.976*	*0.972*	*0.972*	*36*
weighted avg	*0.976*	*0.972*	*0.972*	*36*

Hold-out confusion matrix

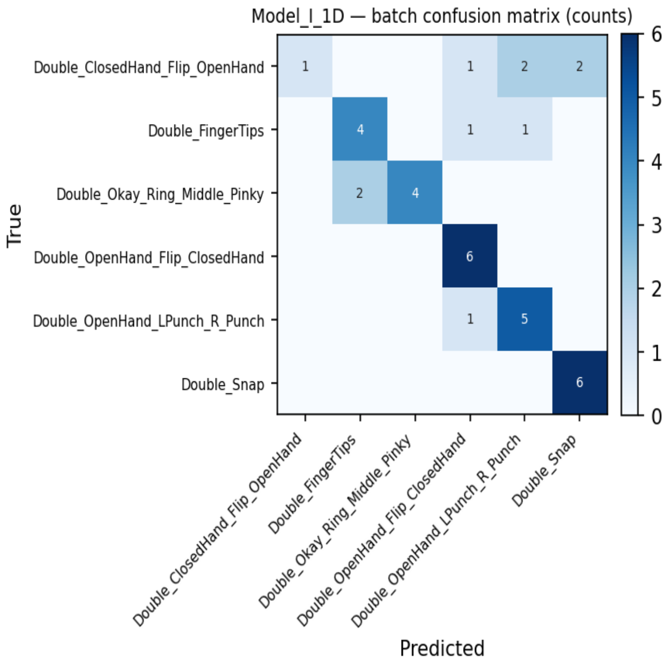


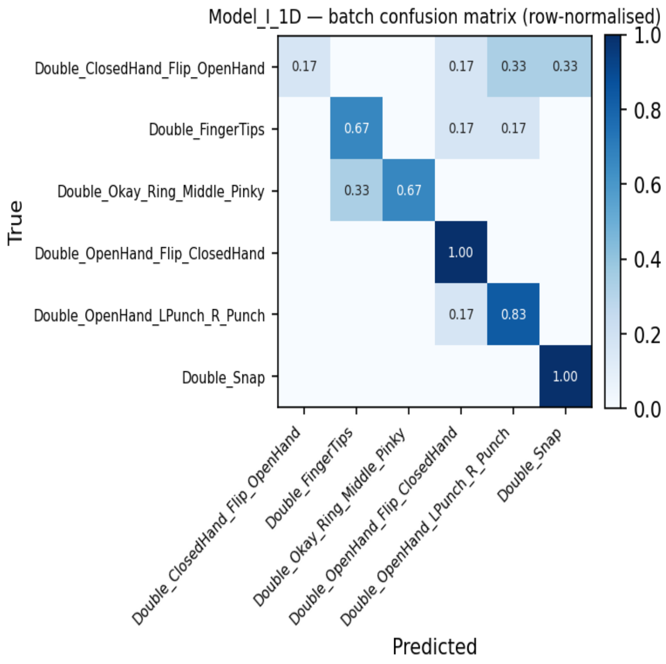


Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	1	6	0.167
Double_FingerTips	4	6	0.667
Double_Okay_Ring_Middle_Pinky	4	6	0.667
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	**26**	**36**	**0.722**

Batch confusion matrix (8_Jestin)


Model_I_1D — batch confusion matrix (counts)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap


Model_I_1D — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap

Variant: Model_II_1D

1-D adaptation of published Model II (CNN+LSTM):
Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

Trainable params	73,926
CV mean ± std	0.9784 ± 0.0157
Hold-out test acc / loss	0.9722 · 0.1889
Batch-test acc	0.7500 (27/36)

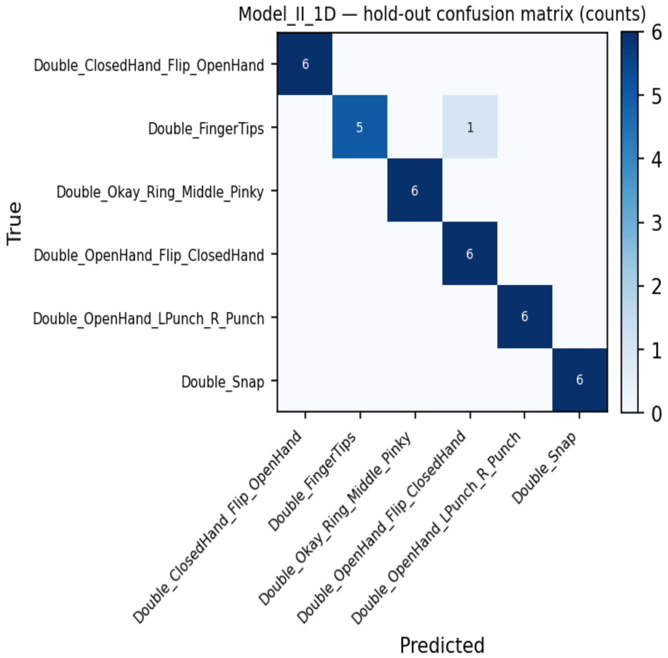
Cross-validation folds

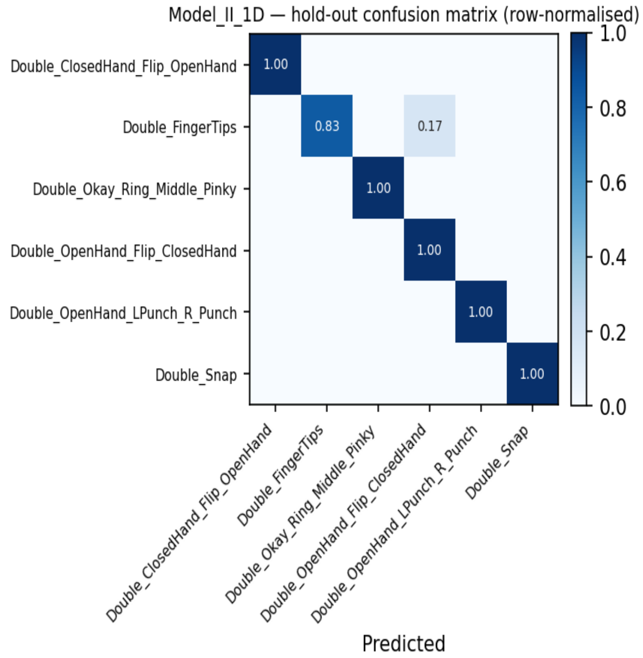
Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1813	65	0.9846	0.9846	0.9846
2	1813	65	0.9538	0.9538	0.9538
3	1813	65	1.0000	1.0000	0.9692
4	1813	65	0.9846	1.0000	0.9692
5	1820	64	0.9688	0.9688	0.9688

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_ClosedHand_Flip_OpenHand	1.000	1.000	1.000	6
Double_FingerTips	1.000	0.833	0.909	6
Double_Okay_Ring_Middle_Pinky	1.000	1.000	1.000	6
Double_OpenHand_Flip_ClosedHand	0.857	1.000	0.923	6
Double_OpenHand_LPunch_R_Punch	1.000	1.000	1.000	6
Double_Snap	1.000	1.000	1.000	6
macro avg	*0.976*	*0.972*	*0.972*	*36*
weighted avg	*0.976*	*0.972*	*0.972*	*36*

Hold-out confusion matrix

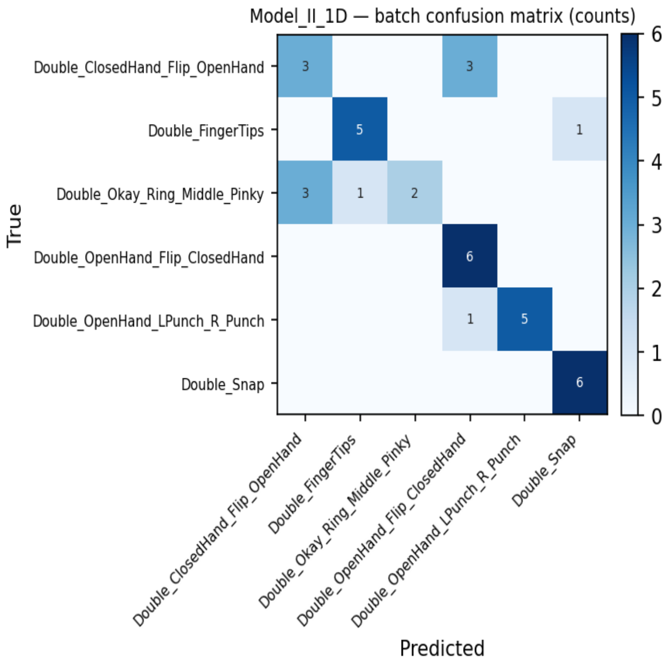


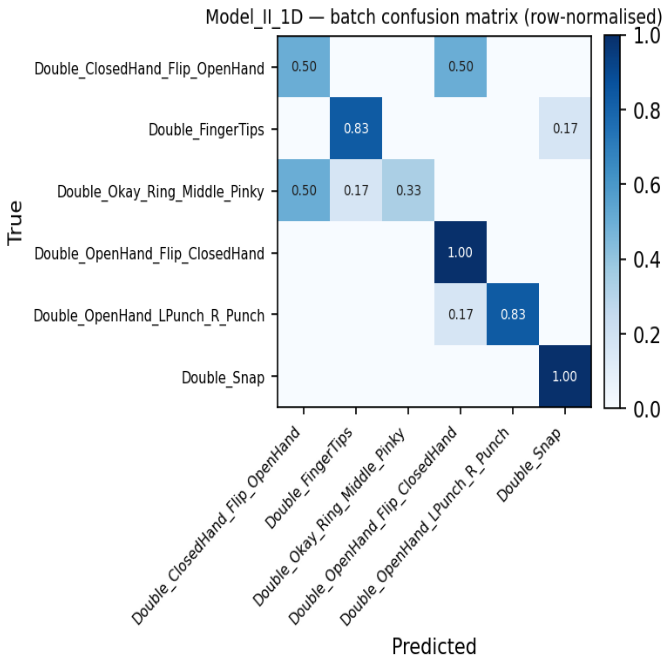


Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_ClosedHand_Flip_OpenHand	3	6	0.500
Double_FingerTips	5	6	0.833
Double_Okay_Ring_Middle_Pinky	2	6	0.333
Double_OpenHand_Flip_ClosedHand	6	6	1.000
Double_OpenHand_LPunch_R_Punch	5	6	0.833
Double_Snap	6	6	1.000
Overall	**27**	**36**	**0.750**

Batch confusion matrix (8_Jestin)


Model_II_1D — batch confusion matrix (counts)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
3
3
5
1
3
1
2
6
1
5
6
0
1
2
3
4
5
6


Model_II_1D — batch confusion matrix (row-normalised)
True
Predicted
Double_ClosedHand_Flip_OpenHand
Double_FingerTips
Double_Okay_Ring_Middle_Pinky
Double_OpenHand_Flip_ClosedHand
Double_OpenHand_LPunch_R_Punch
Double_Snap
0.50
0.50
0.83
0.17
0.50
0.17
0.33
1.00
0.17
0.83
1.00
0.0
0.2
0.4
0.6
0.8
1.0